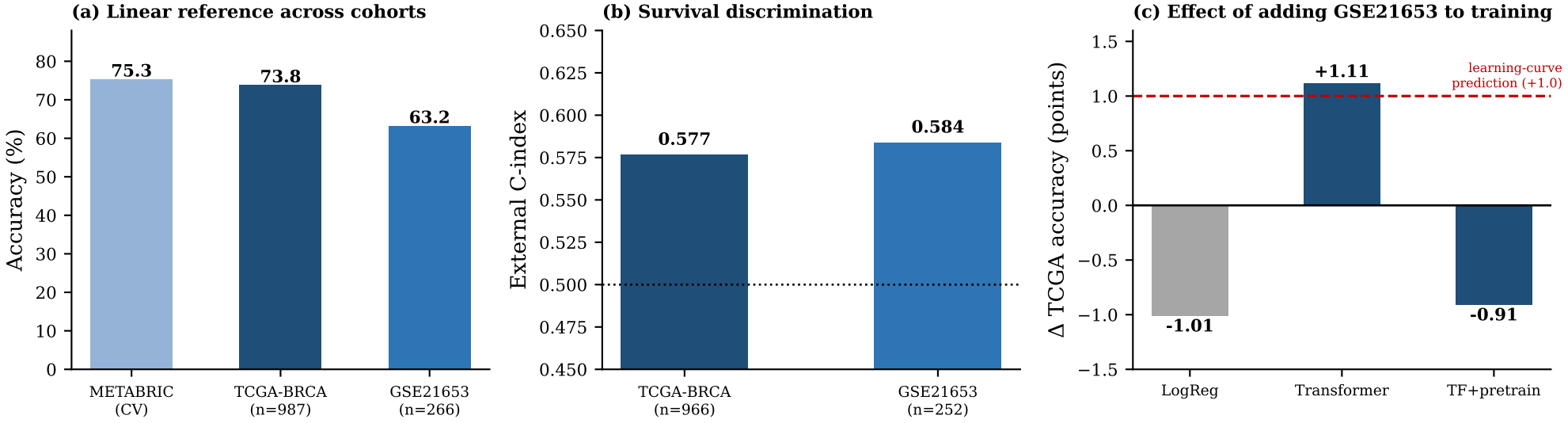
(a) Linear reference across cohorts
Accuracy (%)
0
10
20
30
40
50
60
70
80
75.3
73.8
63.2
METABRIC
(CV)
TCGA-BRCA
(n=987)
GSE21653
(n=266)
(b) Survival discrimination
External C-index
0.450
0.475
0.500
0.525
0.550
0.575
0.600
0.625
0.650
0.577
0.584
TCGA-BRCA
(n=966)
GSE21653
(n=252)
(c) Effect of adding GSE21653 to training
Δ TCGA accuracy (points)
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
learning-curve
prediction (+1.0)
-1.01
+1.11
-0.91
LogReg
Transformer
TF+pretrain